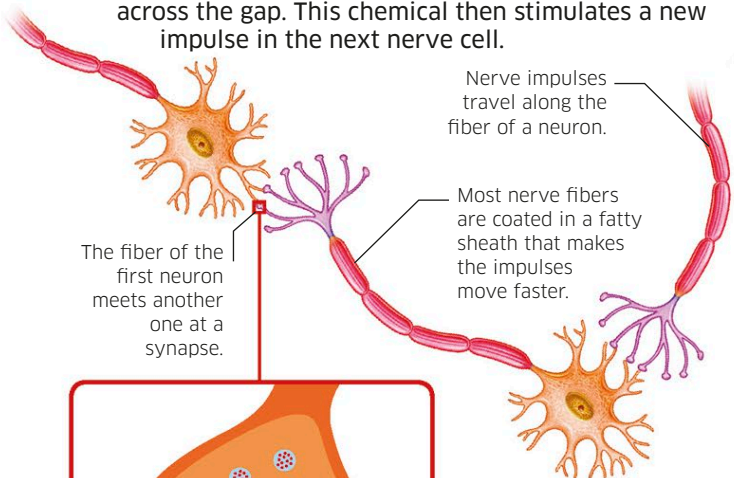


Nerve cells and synapses

Cells of the nervous system have lengthy fibers that can carry electrical signals, called nerve impulses, across long distances. When these signals reach small gaps between cells, called synapses, they trigger the release of a chemical across the gap. This chemical then stimulates a new impulse in the next nerve cell.

**Synapses**

Tiny chemicals called neurotransmitters cross the gap between nerve cells. They are picked up by receptors on the other side.

Responding to surroundings

A gorilla uses its eyes to help sense tasty food, such as wild celery. As they view the food, the eyes send off nerve impulses (electrical signals) to the brain, which then sends instructions to the gorilla's muscles to rip up the plant and eat it.

1 Seeing the plant

Receptors are cells that sense a change in surroundings—called a stimulus. When the receptors in the eye detect light, or “see” the celery, they set off electrical impulses in the nerve cells that are connected to them.

Nerve fibers

Each nerve contains a bundle of microscopic nerve cell fibers. Some nerves carry both sensory and motor fibers; others carry just one or the other.

Impulses

A nerve impulse is a fast-moving spark of electrical activity that runs along the cell membranes of nerve cells (neurons).

6 Hands respond

Parts of the body that move in response to a nerve impulse are called effectors. Muscles are among the most important effectors of an animal's body. When a nerve impulse arrives at a muscle along a motor neuron, it makes the muscle contract (shorten)—in this case to grip and tear the celery.

Nervous system

The speediest body system has cables that carry messages faster than a racing car, and a central control that is smarter than the best computer.

The cables of the nervous system are its nerves, and its control center is the brain. Every moment that the body senses its surroundings, the entire system sends countless electrical impulses through billions of fibers. The nerves trigger muscles to respond, and the brain coordinates all this complex activity.

